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## 摘 要

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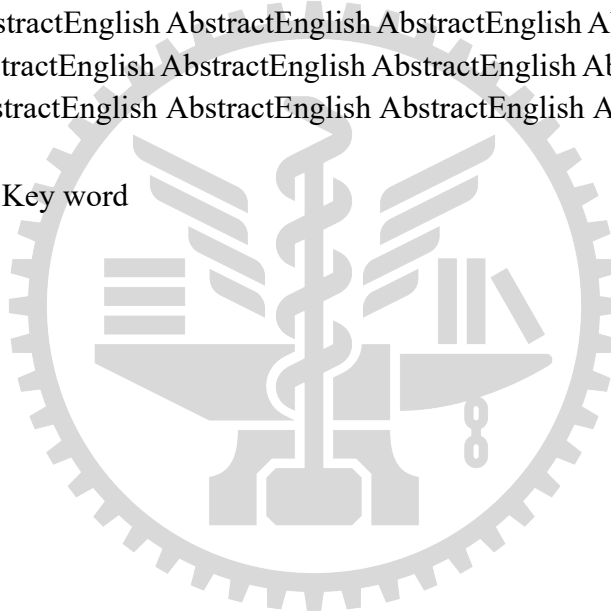
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# Abstract

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# Chapter 1 Introduction

## 1.1 Exciton in Transition Metal Dichalcogenide (TMD) Monolayer

Conventional semiconductors such as diamond, silicon [1], and aluminum arsenide are not suitable for the selective initialization, manipulation, and readout of valley states. This is because they possess multiple inequivalent valleys in their band structures that do not couple distinctly to external optical fields. The situation changed with the discovery of graphene in 2004 [2], which triggered rapid progress in research on two-dimensional (2D) materials. Graphene has a hexagonal honeycomb lattice that places two inequivalent valleys at the  $K$  and  $K'$  points of the hexagonal Brillouin zone. Nevertheless, the absence of a bandgap in graphene allows electrons to be thermally excited between valleys, making valley polarization unstable at room temperature.

In contrast, transition-metal dichalcogenide (TMD) monolayers share a similar honeycomb lattice but possess a finite direct bandgap (see Figure. 1.1) on the order of  $\sim 1\text{--}2\text{ eV}$ , which lies in the visible to near-infrared spectral range. A TMD monolayer consists of two group-VIA chalcogen atoms, such as sulfur (S) or selenium (Se), with a group-VIB transition-metal atom, such as molybdenum (Mo) or tungsten (W), located between them.

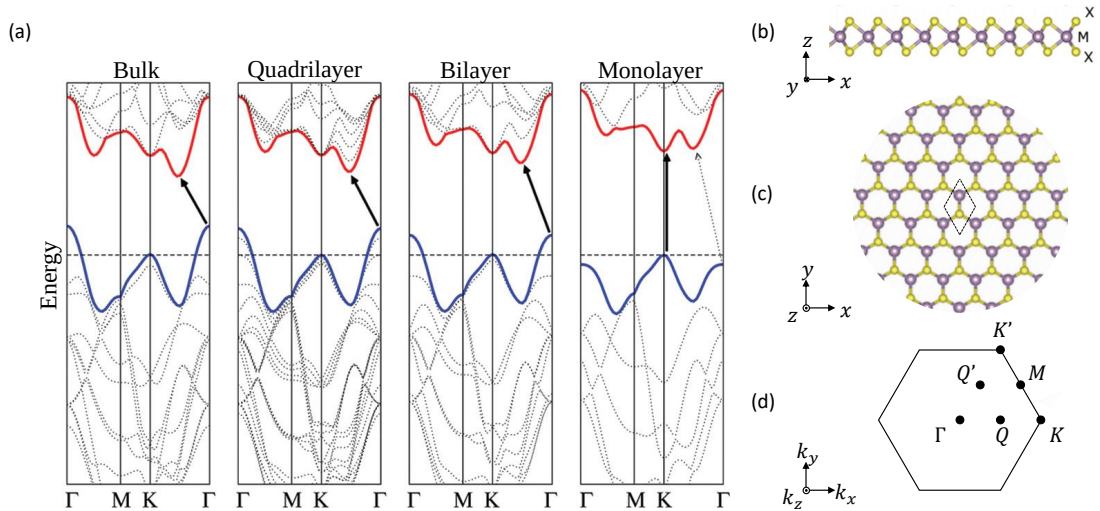


Figure 1.1 : (a) Calculated band structures of MoS<sub>2</sub> with different layers, reproduced from Ref. [3]. (b) Side view and (c) top view of transition-metal dichalcogenide (TMD) monolayer with the unit cell framed by the dashed line, reproduced from Ref. [4]. (d) The first Brillouin zone of TMD monolayer in the reciprocal space and the selected high symmetry points

Under optical excitation, electrons in a TMD monolayer can be promoted from the valence band to the conduction band near the  $K$  and  $K'$  valleys, leaving holes in the valence band. Due to the strong Coulomb interaction in atomically thin materials with weak dielectric screening, the electron-hole pair can bind together to form a quasiparticle known as an exciton. Figure. 1.2 shows that center-of-mass (COM) momentum of exciton is related to momentum of light  $Q_c$  with incident angle  $\theta$  through  $Q = Q_c \sin \theta$ , which gives the idea of the light cone.

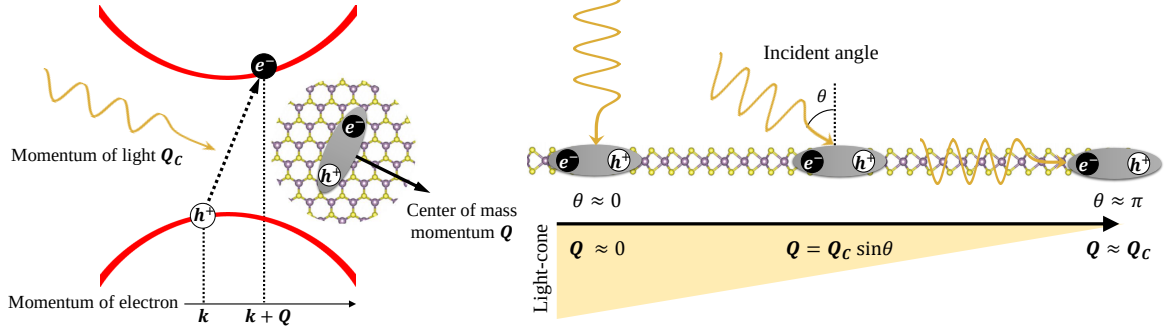


Figure 1.2 : Illustration of a photon carrying momentum  $Q_c$  exciting (leaving) an electron (hole) with momentum  $k + Q$  ( $-k$ ), forming an exciton with center-of-mass (COM) momentum  $Q$ . Schematic of ideal of light cone and relation between COM momentum  $Q$  and  $Q_c$

Moreover, the lack of inversion symmetry in the lattice, together with strong spin-orbit coupling from the heavy transition-metal atoms, gives rise to pronounced spin splitting of the electronic bands. At the  $K/K'$  valleys, this splitting is typically on the order of a few meV in the conduction band and several hundreds of meV in the valence band, as illustrated in Fig. 1.3(a).

The same symmetry considerations also lead to valley-dependent optical selection rules. As shown in Fig. 1.3(b), excitons can be classified according to both momentum and spin selection rules. Only excitons with center-of-mass (COM) momentum  $Q$  within the light cone, i.e.,  $|Q| \lesssim Q_c$ , and with antiparallel spin configurations can recombine radiatively and are classified as bright excitons (BXs). In contrast, excitons that lie outside the light cone or violate the spin selection rules remain optically inactive and are therefore classified as dark excitons (DXs).

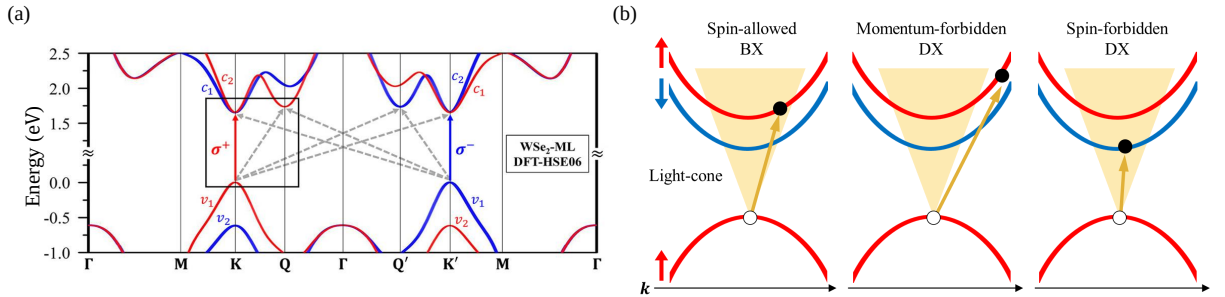


Figure 1.3 : (a) Calculated band structure of WSe<sub>2</sub>, showing spin-up (red) and spin-down (blue) states, along with a schematic illustration of valley-selective optical transitions at the  $K$  and  $K'$  points, which couple to  $\sigma^+$  and  $\sigma^-$  circularly polarized light, respectively, reproduced from Ref. [5]. (b) Schematic configurations of excitonic states, including spin-allowed bright exciton (BX), momentum-forbidden dark exciton (DX), and spin-forbidden DX.

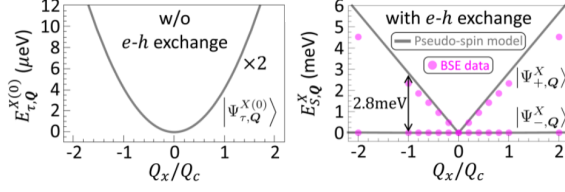


Figure 1.4 : The lowest valley-degenerate (left) and valley-split (right) exciton bands in MoS<sub>2</sub> along the  $Q_x$  direction, reproduced from Ref. [6].

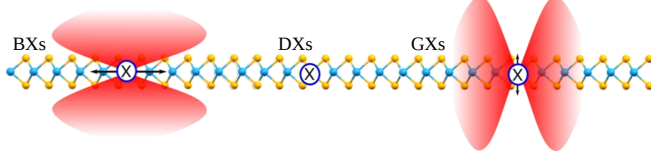


Figure 1.5 : Illustration of the radiation patterns of BXs, DXs, and GXs in neutral WSe<sub>2</sub>, along with the corresponding optical excitation indicated by arrows (reproduced from Ref. [7]).

The finer band structure further enriches the classification of quasiparticles. In particular, the lowest bright exciton band, which is initially twofold degenerate, splits into two linearly dispersing branches when electron–hole exchange interactions are included within a pseudospin model. These branches are commonly referred to as valley-mixed excitons [6].

In W-based TMDs, an additional excitonic state known as the gray exciton (GX) can also emerge. As shown in Fig. 1.5, GXs emit light polarized perpendicular to the two-dimensional plane. Compared with BXs and DXs, GXs exhibit relatively long lifetimes—typically an order of magnitude longer than BXs ( $BX \sim 10$  ps) [7]—while retaining weak but finite optical activity. These properties make GXs particularly useful for studying exciton dynamics and light–matter interactions, and highlight the rich excitonic physics in two-dimensional TMDs.

## 1.2 Spin and Orbital Angular Momentum (SAM and OAM) of Laguerre–Gaussian (LG) beam

Following the theoretical work of Poynting in 1909 [8] and the experimental demonstration by Beth in 1936 [9], it has been established that each photon in circularly polarized light with finite dimensions [10–12] carries a spin angular momentum (SAM) of  $S = \sigma\hbar = \pm\hbar$  along the propagation direction, where  $\sigma$  denotes the photon helicity. In Beth’s experiment, the torque exerted on a suspended half-wave plate by circularly polarized light was measured, demonstrating that spin angular momentum can be transferred to a birefringent medium during propagation (see Fig. 1.6(a)).

In 1992, Allen *et al.* showed that a laser beam in a Laguerre–Gaussian (LG) mode possesses a well-defined orbital angular momentum (OAM), also directed along the propagation axis, with  $L = \ell\hbar$  per photon, where  $\ell \in \mathbb{Z}$  is the azimuthal mode index, commonly referred to as the topological charge [10]. In their experiment, an input Hermite–Gaussian mode, which carries zero OAM, was converted into an LG mode with nonzero OAM. This mode conversion subsequently induced a measurable torque on cylindrical optical elements (see Fig. 1.6(b, c)), establishing that light can also carry OAM associated with the spatial structure of its wavefront. The helical phase structure associated with a quantized amount of  $\ell\hbar$  of OAM is illustrated in Fig. 1.7, while intensified charge-coupled device (ICCD) images show the characteristic doughnut-shaped intensity

distribution.

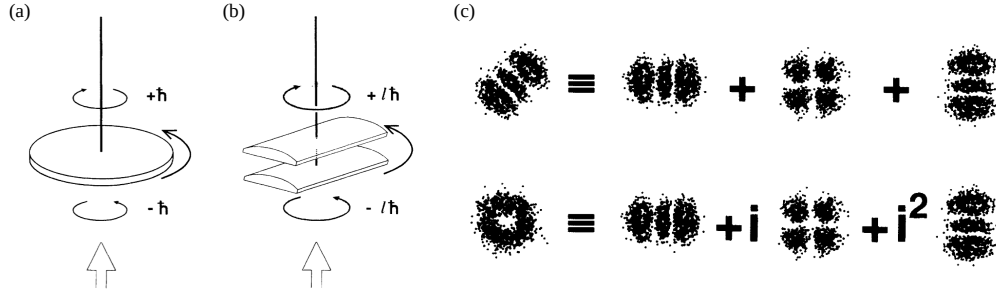


Figure 1.6 : (a) Illustration of the conversion between right-handed and left-handed circularly polarized light generating a torque on a suspended birefringent half-wave plate. (b) Illustration of transforming a Laguerre – Gaussian (LG) mode carrying OAM  $\ell\hbar$  per photon into a mode with  $-\ell\hbar$  per photon, which produces a mechanical torque on suspended cylindrical lenses. (c) Demonstration of generating an LG beam with nonzero  $\ell$  (left side of bottom row) by introducing a phase shift when converting Hermite – Gaussian modes with zero  $\ell$  (left side of top row). All reproduced from Ref. [10].

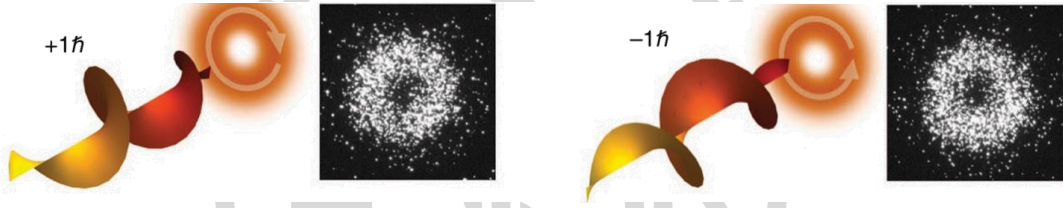


Figure 1.7 : Illustration of twisted light with a helical wavefront, where each photon carries  $\ell\hbar$  of orbital angular momentum. The intensified charge-coupled device camera images show the characteristic doughnut-shaped intensity profile. Reproduced from Ref. [13].

Since then, a wide range of applications of light carrying OAM has been developed. Early studies focused on the transfer of angular momentum from light to matter, where objects experience a mechanical torque due to the transfer of OAM, resulting in rotational motion [14–16]. Later developments have extended to various areas, including optical communications [17–19], quantum key distribution [20, 21], astronomy [22], data storage [23, 24], enhanced sensitivity in imaging [25, 26], and optical tweezers [14, 27]. These developments naturally motivate the investigation of how structured light carrying orbital angular momentum interacts with electronic excitations in low-dimensional materials.

### 1.3 SAM and OAM coupling in LG beam

In free space, coupling between SAM and OAM has been shown to occur in non-paraxial (tightly focused) light. Along the propagation direction, SAM can be partially converted into OAM as the focusing angle increases [28]. This effect can be understood by considering the reduction of the spin component along the beam axis induced by a focusing lens (see Fig. 1.8).

The initial spin  $s_0$  is aligned with the propagation direction. After focusing, however, the spin is no longer parallel to the beam axis; instead, the final spin  $s_1$  is tilted, such that only its longitudinal component  $s_z$  contributes to the total spin angular momentum of the beam. Since the maximum magnitude of the spin angular momentum is  $s\hbar$  per photon, the total spin after focusing must be reduced. However, the total angular momentum is conserved during the focusing process. Consequently, the reduction in SAM must be compensated by the generation of orbital angular momentum, leading to a transfer from SAM to OAM [29].

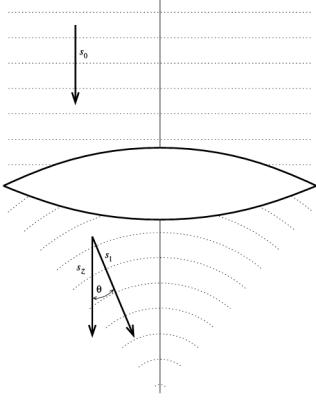


Figure 1.8 : Illustration of a beam focused by a lens, showing the variation of SAM before and after the lens along the propagation direction. Reproduced from Ref. [29].

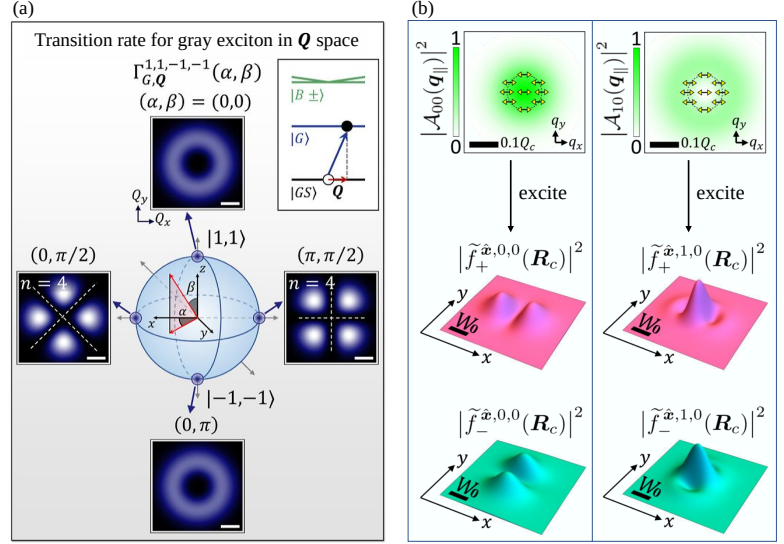


Figure 1.9 : (a) Transition-rate distribution for gray excitons in momentum space, reproduced from Ref. [30]. (b) Distribution of the incident light and the excited BX wave-packet states, reproduced from Ref. [6].

It should be emphasized that the simple proportionality between spin and orbital angular momentum with the optical helicity  $\sigma$  and topological charge  $\ell$  holds strictly only within the paraxial approximation [31, 32], which often neglects the longitudinal component along the propagation direction. However, it is precisely this longitudinal component that can couple to GXs. More importantly, this neglected component already contains two distinct types of spin-orbit interaction (SOI).

In 2024, Oscar Javier Gomez Sanchez *et al.* demonstrated that one type of SOI, arising from the phase of the longitudinal component, enables selective excitation of GXs through the superposition of two twisted light beams, forming a vector vortex beam (VVB) (see Fig. 1.9(a)) [30]. However, the role of the combined SAM and OAM contribution is not explicitly clarified in their formulation. On the other hand, in 2022, Guan-Hao Peng *et al.* demonstrated the photoexcitation of LG beams for valley-mixed BXs within a pseudospin model, suggesting a connection to excitonic topology [6] (see Fig. 1.9(b)). However, the global phase of the valley-mixed excitons may influence the resulting excited wave-packet profile.

In this work, we extend the studies of Gomez Sanchez *et al.* and Peng *et al.* to investigate the interplay between spin and orbital angular momentum of light in determining the transition rate and wave-packet distribution for various excitonic states.

# Chapter 2 SAM and OAM in LG beam

## 2.1 Gauge Restriction in LG Beam

The derivation of Laguerre–Gaussian (LG) beams involves a series of approximations, among which the paraxial approximation plays a central role. Importantly, the wave equation obtained under the paraxial approximation is not independent of the choice of gauge; rather, it is implicitly constrained by the adopted gauge condition.

In this section, we examine how gauge choice restricts the formulation of wave equations. We begin by introducing the wave equations under different gauge conditions, and then analyze the compatibility of the paraxial approximation. This discussion clarifies why certain gauges are more suitable than others for describing paraxial optical fields.

### 2.1.1 Gauge Freedom in Wave Equations

Maxwell's equations describe how the **electric field**  $\mathbf{E}(\mathbf{r}, t)$  and the **magnetic field**  $\mathbf{B}(\mathbf{r}, t)$  evolve in space and time. These equations form the fundamental basis for describing electromagnetic waves, including Laguerre–Gaussian beams. In **free space**, the charge density  $\rho$  and current density  $\mathbf{J}$  vanish. In this case, Maxwell's equations reduce to four homogeneous differential equations involving the vacuum permittivity  $\epsilon_0$ , the vacuum permeability  $\mu_0$ , and the speed of light in vacuum  $c = 1/\sqrt{\mu_0\epsilon_0}$ .

Within this electrodynamics framework, Maxwell's equations can be rewritten as two coupled wave equations for the **scalar and vector potentials**,  $\phi(\mathbf{r}, t)$  and  $\mathbf{A}(\mathbf{r}, t)$ , derived in Appendix.1 as

$$\nabla^2 \phi(\mathbf{r}, t) + \frac{\partial}{\partial t} (\nabla \cdot \mathbf{A}(\mathbf{r}, t)) = 0, \quad (2.1)$$

$$\nabla^2 \mathbf{A}(\mathbf{r}, t) - \frac{1}{c^2} \frac{\partial^2}{\partial t^2} \mathbf{A}(\mathbf{r}, t) - \nabla \left( \nabla \cdot \mathbf{A}(\mathbf{r}, t) + \frac{1}{c^2} \frac{\partial}{\partial t} \phi(\mathbf{r}, t) \right) = 0. \quad (2.2)$$

These equations contain the same physical information as the original Maxwell equations, but are expressed in terms of the scalar and vector potentials.

Further simplification of Eqs. (2.1) and (2.2) can be achieved by exploiting the gauge freedom of the potentials. Below, we introduce two commonly used gauges: the **Coulomb gauge** and the **Lorenz gauge**. Superscripts  $C$  and  $L$  are used to distinguish quantities defined in the

Coulomb and Lorenz gauges, respectively. The gauge conditions are given by

$$\text{Coulomb gauge condition : } \nabla \cdot \mathbf{A}^C(\mathbf{r}, t) = 0, \quad (2.3)$$

$$\text{Lorenz gauge condition : } \nabla \cdot \mathbf{A}^L(\mathbf{r}, t) = -\frac{1}{c^2} \frac{\partial}{\partial t} \phi^L(\mathbf{r}, t). \quad (2.4)$$

Both gauges simplify the wave equations. In the Coulomb gauge, they reduce to

$$\nabla^2 \phi^C(\mathbf{r}, t) = 0, \text{ and} \quad (2.5)$$

$$\nabla^2 \mathbf{A}^C(\mathbf{r}, t) - \frac{1}{c^2} \frac{\partial^2}{\partial t^2} \mathbf{A}^C(\mathbf{r}, t) = \frac{1}{c^2} \frac{\partial}{\partial t} \nabla \phi^C(\mathbf{r}, t) = 0, \quad (2.6)$$

where the last equality follows from Eq. (2.5).

In the Lorenz gauge, Eqs. (2.1) and (2.2) become

$$\nabla^2 \phi^L(\mathbf{r}, t) - \frac{1}{c^2} \frac{\partial^2}{\partial t^2} \phi^L(\mathbf{r}, t) = 0, \text{ and} \quad (2.7)$$

$$\nabla^2 \mathbf{A}^L(\mathbf{r}, t) - \frac{1}{c^2} \frac{\partial^2}{\partial t^2} \mathbf{A}^L(\mathbf{r}, t) = 0. \quad (2.8)$$

In general, one may choose either gauge and proceed with the derivation; however, in the next section we will show that the Lorenz gauge is the most suitable for our purposes. The physical field  $\mathbf{B}(\mathbf{r}, t)$  and  $\mathbf{E}(\mathbf{r}, t)$  can be related to the vector potentials defined in different gauges through Eq. (A.1.5) and

$$\mathbf{E}(\mathbf{r}) = i\omega \mathbf{A}^C(\mathbf{r}) = i\omega \left( \mathbf{A}^L(\mathbf{r}) + \frac{\nabla (\nabla \cdot \mathbf{A}^L(\mathbf{r}))}{q_0^2} \right). \quad (2.9)$$

The derivation of this relation assumes a **monochromatic wave** ansatz. Specifically, the fields and potentials are written as  $\mathbf{E}(\mathbf{r}, t) = \text{Re} [\mathbf{E}(\mathbf{r}) e^{-i\omega t}]$ ,  $\phi^L(\mathbf{r}, t) = \text{Re} [\phi^L(\mathbf{r}) e^{-i\omega t}]$ ,  $\mathbf{A}^C(\mathbf{r}, t) = \text{Re} [\mathbf{A}^C(\mathbf{r}) e^{-i\omega t}]$ , and  $\mathbf{A}^L(\mathbf{r}, t) = \text{Re} [\mathbf{A}^L(\mathbf{r}) e^{-i\omega t}]$ , where  $\omega$  is the angular frequency. Substituting these expressions into Eq. (A.1.7) and applying the corresponding gauge conditions leads to Eq. (2.9). Here  $q_0 = \omega/c$  denotes the wavenumber of light in free space.

### 2.1.2 Incompatibility of Gauge Freedom with the Paraxial Approximation

In principle, the vector potential formulation of LG beams can be derived starting from either Eq. (2.6) or Eq. (2.8). However, only the Lorenz gauge remains fully consistent throughout the derivation, whereas the Coulomb gauge condition becomes incompatible with the paraxial wave ansatz. For this reason, we adopt the Lorenz gauge in the following derivation for simplicity and consistency, and verify this choice at the end.

Before applying the paraxial approximation, one obtains the scalar Helmholtz equation through both monochromatic ansatz and transverse wave ansatz within the Lorenz gauge (see

Sec. Appendix.2):

$$\nabla^2 A^L(\mathbf{r}) + q_0^2 A^L(\mathbf{r}) = 0. \quad (2.10)$$

The **paraxial approximation** assumes that the rays have only small inclinations with respect to the optical axis in geometric optics. In wave optics, this corresponds to an angular spectrum dominated by plane-wave components whose wavevectors are close to the propagation direction. Here, we choose the propagation direction along the  $z$  axis, such that the dominant component of the wavevector  $\mathbf{q}_0$  lies along  $\hat{z}$ . Under this approximation, we introduce the **paraxial wave ansatz**,  $A^L(\mathbf{r}) = U(\mathbf{r})e^{iq_0 z}$ , and substitute it into Eq. (A.2.2), which yields

$$\nabla_{\parallel}^2 U(\mathbf{r}) + \frac{\partial^2}{\partial z^2} U(\mathbf{r}) + 2iq_0 \frac{\partial}{\partial z} U(\mathbf{r}) = 0. \quad (2.11)$$

Here,  $\nabla_{\parallel}^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}$  denotes the transverse Laplacian. The paraxial wave ansatz assumes that the **envelope function**  $U(\mathbf{r})$  **varies slowly along the  $z$  direction** compared to the rapid phase variation of  $e^{iq_0 z}$ . Consequently, the paraxial approximation implies the following relations:

$$\left| \frac{\partial^2}{\partial z^2} U(\mathbf{r}) \right| \ll \left| \nabla_{\parallel}^2 U(\mathbf{r}) \right|, \quad \left| \frac{\partial^2}{\partial z^2} U(\mathbf{r}) \right| \ll q_0 \left| \frac{\partial}{\partial z} U(\mathbf{r}) \right|.$$

Neglecting the second-order derivative with respect to  $z$  then leads to the **paraxial Helmholtz equation** [33]

$$\nabla_{\parallel}^2 U(\mathbf{r}) + 2iq_0 \frac{\partial}{\partial z} U(\mathbf{r}) = 0. \quad (2.12)$$

We now clarify why the Coulomb gauge is not suitable in this derivation. The paraxial wave ansatz generally does not satisfy the Coulomb gauge condition (Eq. (2.3)), which requires

$$\nabla \cdot \mathbf{A}^C(\mathbf{r}, t) = \hat{\varepsilon} \cdot \nabla A^C(\mathbf{r}, t) = 0. \quad (2.13)$$

For a transverse wave with polarization  $\hat{\varepsilon}$  confined to the  $x$ - $y$  plane, this condition reduces to

$$\hat{\varepsilon} \cdot \nabla_{\parallel} A^C(\mathbf{r}, t) = 0, \quad (2.14)$$

where  $\nabla_{\parallel} = \frac{\partial}{\partial x} \hat{x} + \frac{\partial}{\partial y} \hat{y}$ .

This condition constrains the transverse spatial variation of  $A^C(\mathbf{r}, t)$  along the polarization direction, which is incompatible with the assumed slowly varying envelope  $U(\mathbf{r})$  [33]. In other words, the Coulomb gauge imposes an additional constraint that conflicts with the paraxial ansatz. To satisfy the divergence-free condition, a nonzero longitudinal ( $z$ ) component of the vector potential must be introduced, going beyond the simple transverse ansatz. Therefore, the Lorenz gauge provides a more natural and consistent framework for deriving the paraxial Helmholtz equation and Laguerre–Gaussian beam solutions.

## 2.2 SAM and OAM Interaction (SOI) in LG Beam

Within the paraxial approximation, the optical helicity  $\sigma$  and the topological charge  $\ell$  are directly proportional to the spin and orbital angular momentum of light, respectively [31, 32]. This correspondence provides a convenient framework for analyzing the angular momentum properties of Laguerre–Gaussian (LG) beams.

Moreover, when the vector-potential formulation of LG beams is expressed in the Coulomb gauge, an SAM and OAM interaction (SOI) can emerge through the additional longitudinal component of the field, giving rise to terms proportional to  $\sigma\ell$  and  $\sigma + \ell$ . Focusing on the Coulomb gauge is further motivated by the theory of light–matter interaction, in which the vector potential is naturally formulated in both the Coulomb gauge and a plane-wave basis. In this representation, approximately interpret the in-plane wavevector as a vector in reciprocal space enable a connection to theory of matter.

In the following section, we first introduce the LG beam solution in both real-space and angular-spectrum representations. We then perform a gauge transformation to reveal the SOI.

### 2.2.1 LG Beam in two representations

Under the **long Rayleigh-length range** [34], the LG beam in real space can be solved from Eq. (2.12) within cylindrical coordinates as

$$A_{\mathbf{q}_0}^{\hat{\epsilon}, \ell, p; L}(\mathbf{r}) = \hat{\epsilon} A_0^{\text{LG}} u_{\ell p}(\mathbf{r}_{\parallel}) e^{iq_0 z}, \quad (2.15)$$

where the position vector is written as  $\mathbf{r} = (\rho, \phi, z) = (\mathbf{r}_{\parallel}, z)$ . Here  $\mathbf{r}_{\parallel}$  denotes the in-plane position vector consisting of the radial coordinate  $\rho$  and the azimuthal angle  $\phi$ . The integer  $\ell \in \mathbb{Z}$  is the **topological charge**, and the integer  $p \geq 0$  is the radial mode index. The vector  $\mathbf{q}_0 = q_0 \hat{z}$  denotes the wavevector along the propagation direction, and  $A_0^{\text{LG}}$  is a constant amplitude.

The transverse spatial distribution is expressed as

$$U(\rho, \phi, z) \approx u_{\ell p}(\mathbf{r}_{\parallel}) = f_{\ell p}(\rho) e^{i\ell\phi}, \quad (2.16)$$

where the radial function is given by  $f_{\ell p}(\rho) = \sqrt{\frac{2p!}{\pi(|\ell|+p)!}} L_p^{|\ell|} \left( \frac{2\rho^2}{W_0^2} \right) \left( \frac{\sqrt{2}\rho}{W_0} \right)^{|\ell|} \exp \left( -\frac{\rho^2}{W_0^2} \right)$ . Here  $L_p^{|\ell|}(x)$  is the **generalized Laguerre polynomial**, defined as  $L_p^{|\ell|}(x) = \sum_{m=0}^p (-1)^m \frac{(|\ell|+p)!}{(p-m)! (|\ell|+m)! m!} x^m$ , with  $x = \frac{2\rho^2}{W_0^2}$ . The parameter  $W_0$  denotes the beam waist of the LG mode, while the cylindrical coordinates are related to Cartesian coordinates through  $\rho = \sqrt{x^2 + y^2}$  and  $\phi = \tan^{-1}(y/x)$ . This form is consistent with that obtained by Guan-Hao Peng *et al.* in Ref. [6].

Figure. 2.1 shows that as  $\ell$  and/or  $p$  increase,  $u_{\ell p}(\mathbf{r}_{\parallel})$  develops a broader doughnut-shaped intensity profile, accompanied by an increasing number of radial nodes along the radial direction. The phase varies continuously with the azimuthal angle  $\phi$ , reflecting the helical wavefront associated with orbital angular momentum. In particular, the phase winds by  $2\pi\ell$  around the

beam center. The beam waist is chosen as  $W_0 = 1.5 \mu\text{m}$ , which sets the spatial scale of the beam and determines the radial extent of the intensity distribution.

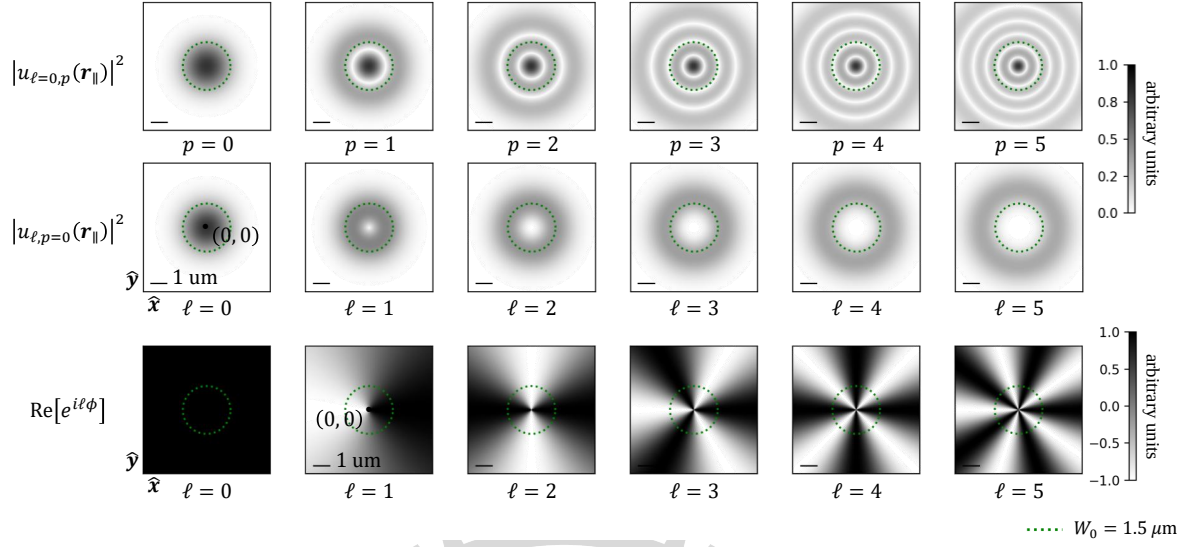


Figure 2.1 : Squared magnitude (intensity) and phase distribution of  $u_{\ell,p}(\mathbf{r}_{\parallel})$ . The green dotted line denotes the beam waist  $W_0$ . The scale bar indicates the length of 1  $\mu\text{m}$

The vector potential of LG beams in a plane-wave basis can be related to that in real space as

$$\mathbf{A}(\mathbf{r}) = \sum_{\mathbf{q}} \mathcal{A}(\mathbf{q}) e^{i\mathbf{q} \cdot \mathbf{r}}, \quad (2.17)$$

where  $\mathcal{A}(\mathbf{q})$  is the complex amplitude associated with each plane-wave component  $e^{i\mathbf{q} \cdot \mathbf{r}}$ . The LG beam in Eq. (2.15) can be written in this form through a defined **two-dimensional Fourier transformation** and its corresponding inverse transformation,

$$u_{\ell p}(\mathbf{r}_{\parallel}) = \frac{1}{(2\pi)^2} \int d^2 \mathbf{q}_{\parallel} F_{\ell p}^{\text{LG}}(\mathbf{q}_{\parallel}) e^{i\mathbf{q}_{\parallel} \cdot \mathbf{r}_{\parallel}}, \quad \longleftrightarrow \quad F_{\ell p}^{\text{LG}}(\mathbf{q}_{\parallel}) = \int d^2 \mathbf{r}_{\parallel} u_{\ell p}(\mathbf{r}_{\parallel}) e^{-i\mathbf{q}_{\parallel} \cdot \mathbf{r}_{\parallel}}. \quad (2.18)$$

For applications to two-dimensional materials, the in-plane wavevector  $\mathbf{q}_{\parallel}$  can be approximately regarded as a vector in reciprocal space. Under this approximation, Eq. (2.18) can be discretized, and the Eq. (2.17) becomes

$$\mathbf{A}_{\mathbf{q}_0}^{\hat{\varepsilon}, \ell, p; L}(\mathbf{r}) = \hat{\varepsilon} \sum_{\mathbf{q}_{\parallel}} \mathcal{A}_{\ell p}^L(\mathbf{q}_{\parallel}) e^{i\mathbf{q}_{\parallel} \cdot \mathbf{r}_{\parallel}} e^{i\mathbf{q}_0 \cdot \mathbf{z}}. \quad (2.19)$$

Here the in-plane wavevector is written as  $\mathbf{q}_{\parallel} = (q_{\parallel}, \phi_{\mathbf{q}_{\parallel}})$ . Since the spatial distribution has the form  $u_{\ell p}(\mathbf{r}_{\parallel}) = f_{\ell p}(\rho) e^{i\ell\phi}$ , the angular spectrum amplitude can be expressed analytically as

$$\mathcal{A}_{\ell p}^L(\mathbf{q}_{\parallel}) = \tilde{F}_{\ell p}(q_{\parallel}) e^{i\ell\phi_{\mathbf{q}_{\parallel}}}, \quad (2.20)$$

where  $\tilde{F}_{\ell p}(q_{\parallel}) = \frac{A_0^{\text{LG}}}{\Omega} (-1)^{|\ell|} 2\pi \mathcal{H}_{|\ell|}[f_{\ell p}(\rho)]$ , and  $\mathcal{H}_{|\ell|}[f_{\ell p}(\rho)] = \int_0^{\infty} d\rho \rho f_{\ell p}(\rho) J_{|\ell|}(q_{\parallel}\rho)$  is the **Hankel transform** of order  $|\ell|$ , with  $J_{|\ell|}(q_{\parallel}\rho)$  being the **Bessel function of the first kind**. Here

$\Omega$  denotes the area of the two-dimensional system.

In this study, we focus on the fundamental radial mode  $p = 0$ . In real space, the vector potential takes the explicit form

$$\mathbf{A}_{\mathbf{q}_0}^{\hat{\varepsilon}, \ell, p=0; L}(\mathbf{r}) = \hat{\varepsilon} \frac{A_0^{\text{LG}}}{\sqrt{|\ell|!} \pi/2} \left( \frac{\sqrt{2}\rho}{W_0} \right)^{|\ell|} \exp\left(-\frac{\rho^2}{W_0^2}\right) \exp[i(\ell\phi + q_0 z)], \quad (2.21)$$

whereas in the angular spectrum representation the amplitude becomes

$$\mathcal{A}_{\ell, p=0}^L(\mathbf{q}_{\parallel}) = \left( \frac{\pi A_0^{\text{LG}} W_0^2 (-1)^{|\ell|}}{\Omega \sqrt{|\ell|!} \pi/2} \right) \left[ \left( \frac{q_{\parallel} W_0}{\sqrt{2}} \right)^{|\ell|} \exp\left(-\frac{q_{\parallel}^2 W_0^2}{4}\right) \right] \exp(i\ell\phi_{\mathbf{q}_{\parallel}}). \quad (2.22)$$

This form is consistent with that obtained by Guan-Hao Peng *et al.* in Ref. [6].

Figure 2.2 shows that  $\mathcal{A}_{\ell, p=0}^L(\mathbf{q}_{\parallel})$  exhibits an intensity profile analogous to that of  $u_{\ell, p=0}(\mathbf{r}_{\parallel})$ , reflecting the characteristic doughnut-like structure in momentum space. This correspondence follows directly from the Fourier transform relation between real space and angular spectrum representations of the Laguerre–Gaussian mode. In particular, the doughnut-shaped distribution in momentum space indicates that the dominant in-plane wavevectors lie within a finite range determined by the beam waist  $W_0$ .

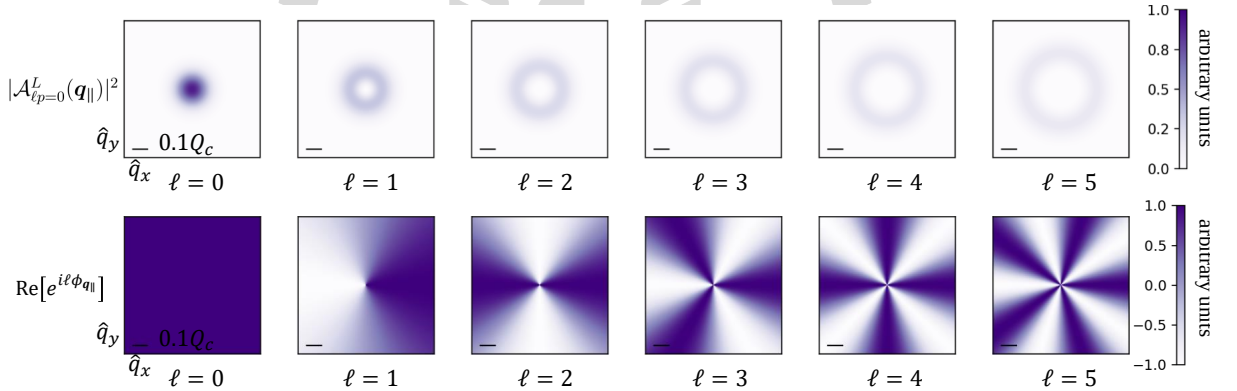


Figure 2.2 : Squared magnitude and phase distribution of  $\mathcal{A}_{\ell, p=0}^L(\mathbf{q}_{\parallel})$ . The scale bar indicates the considered light-cone range for WSe<sub>2</sub>-monolayers,  $Q_c = 8.6 \mu\text{m}^{-1}$ .

## 2.2.2 OAM and SAM Interaction in the Nonzero $z$ Component

In the previous section, Eq. (2.9) was introduced to relate the vector potentials defined in the two gauges to the physical fields. Using this relation, the Coulomb-gauge vector potential can be written as

$$\mathbf{A}_{\mathbf{q}_0}^{\hat{\varepsilon}, \ell, p; C}(\mathbf{r}) = \mathbf{A}_{\mathbf{q}_0}^{\hat{\varepsilon}, \ell, p; L}(\mathbf{r}) + \frac{\nabla (\nabla \cdot \mathbf{A}_{\mathbf{q}_0}^{\hat{\varepsilon}, \ell, p; L}(\mathbf{r}))}{q_0^2}. \quad (2.23)$$

To explicitly incorporate the spin angular momentum of light, the polarization vector is defined as  $\hat{\varepsilon} = \frac{1}{\sqrt{2}} (\hat{x} + i\sigma\hat{y})$ , where the photon helicity  $\sigma = \pm 1$  denotes circular polarization. For LG beams in the fundamental radial mode  $p = 0$ , the vector potential in real space under

the Coulomb gauge is obtained as

$$\mathbf{A}_{q_0}^{\hat{\varepsilon}, \ell, p=0; C}(\mathbf{r}) \approx A_0^{\text{LG}} \left[ f_{\ell, p=0}(\rho) e^{i\ell\phi} \hat{\varepsilon} + \frac{i}{\sqrt{2}q_0\rho} \left( (|\ell| + \sigma\ell) - \frac{2\rho^2}{W_0^2} \right) f_{\ell, p=0}(\rho) e^{i(\sigma+\ell)\phi} \hat{z} \right] e^{iq_0 z}. \quad (2.24)$$

This form is consistent with that obtained by Oscar Javier Gomez Sanchez *et al.* in Ref. [30]. In general, the second term in Eq. (2.23) introduces corrections to all spatial components of the vector potential. However, under the paraxial approximation, these corrections are suppressed. Therefore, we retain only the longitudinal ( $z$ ) component as the leading correction.

The longitudinal component in Eq. (2.24) reveals two distinct types of SOI: a  $\sigma\ell$  dependence in the amplitude and a  $\sigma + \ell$  dependence in the phase. Figure 2.3 shows that the magnitude (absolute value) of this longitudinal component is significantly weaker than that of the transverse components. The  $\sigma\ell$  dependence manifests as an asymmetry between cases with the same  $\ell$  but opposite helicities  $\sigma$ .

It is worth noting that the longitudinal term is weighted by a factor proportional to  $(q_0 W_0^2)^{-1}$ , which is inversely related to the Rayleigh range ( $\frac{1}{2}q_0 W_0^2$ ). This indicates that a smaller beam waist  $W_0$  leads to a stronger longitudinal component, corresponding to a shorter Rayleigh range and tighter focusing, as shown in the bottom row of Fig. 2.3. The beam waist  $W_0$  is chosen to be  $1.5 \mu\text{m}$  and  $0.5 \mu\text{m}$ , representing a typical paraxial beam and a case near the boundary of the paraxial regime, respectively [35–37]. This demonstrates that the longitudinal component, although small, remains finite even within the paraxial regime.

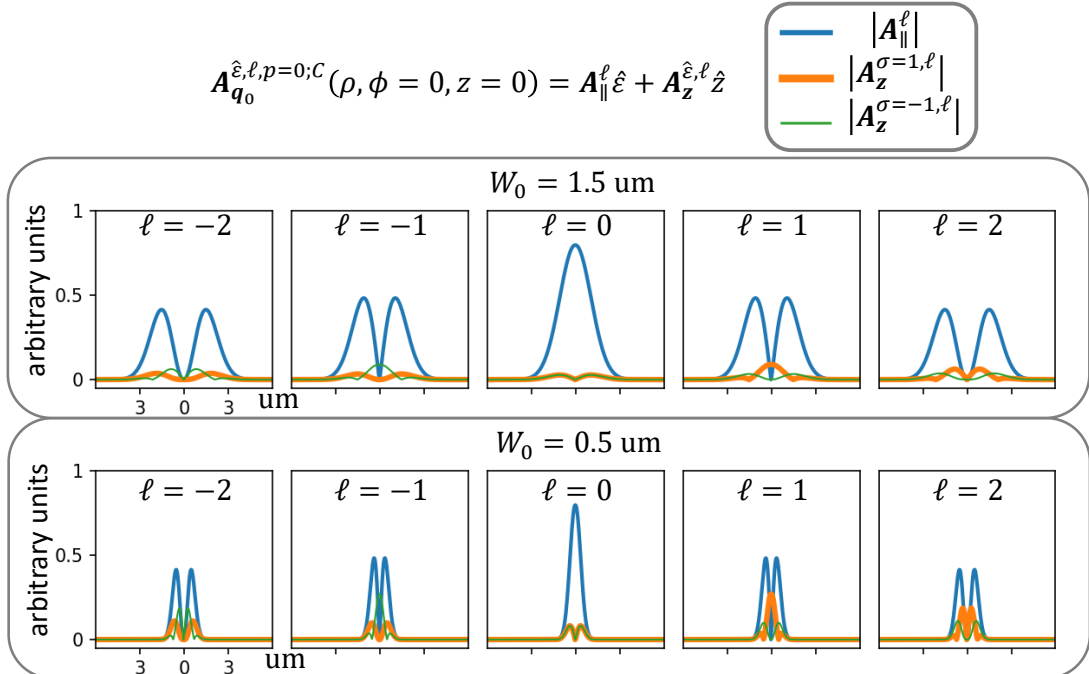


Figure 2.3 : Radial dependence of  $\mathbf{A}_{q_0}^{\hat{\varepsilon}, \ell, p=0; C}(\mathbf{r})$  for different components and helicities (absolute value). The  $\sigma\ell$  term leads to asymmetry between cases with the same  $\ell$  in the longitudinal component (a wavelength of  $532 \text{ nm}$  is assumed).

As for the  $\sigma + \ell$  dependence in the phase term, it is retained in the angular-spectrum representation and will be discussed below. By combining Eq. (2.23) with the two-dimensional Fourier transformation defined in Coulomb-gauge vector potential can be written as

$$\mathcal{A}_{q_0}^{\hat{\varepsilon}, \ell, p=0; C}(\mathbf{r}) = \frac{1}{(2\pi)^2} \int d^2 \mathbf{q}_{\parallel} \mathcal{A}_{\ell p}^C(\mathbf{q}_{\parallel}) e^{i \mathbf{q}_{\parallel} \cdot \mathbf{r}_{\parallel}} e^{i q_0 z}. \quad (2.25)$$

The corresponding amplitude in the plane-wave basis is obtained as

$$\mathcal{A}_{\ell p}^C(\mathbf{q}_{\parallel}) = \left[ \hat{\varepsilon} - \left( \frac{|\mathbf{q}|}{q_0} \right)^2 (\hat{\varepsilon} \cdot \hat{\mathbf{q}}_{\parallel}) \hat{\mathbf{q}} \right] \mathcal{A}_{\ell p}^L(\mathbf{q}_{\parallel}) \quad (2.26)$$

$$= \left[ \hat{\varepsilon} - \left( \frac{|\mathbf{q}|}{q_0} \right)^2 \frac{\sin \theta_q}{\sqrt{2}} e^{i \sigma \phi_{q_{\parallel}}} \hat{\mathbf{q}} \right] \mathcal{A}_{\ell p}^L(\mathbf{q}_{\parallel}) \quad (2.27)$$

$$\approx \left[ \hat{\varepsilon} - \frac{\sin \theta_q}{\sqrt{2}} e^{i \sigma \phi_{q_{\parallel}}} \hat{\mathbf{q}} \right] \mathcal{A}_{\ell p}^L(\mathbf{q}_{\parallel}), \quad (2.28)$$

where the last approximation follows from the paraxial condition  $\frac{q_0}{|q|} = \cos \theta_q \approx 1$ . The unit vector is given by  $\hat{\mathbf{q}} = \sin \theta_q (\cos \phi_{q_{\parallel}} \hat{\mathbf{x}} + \sin \phi_{q_{\parallel}} \hat{\mathbf{y}}) + \cos \theta_q \hat{\mathbf{z}}$ , and  $\sin \theta_q = \frac{q_{\parallel}}{\sqrt{q_{\parallel}^2 + q_0^2}}$ . Substituting the Lorenz-gauge amplitude obtained previously, the Coulomb-gauge vector potential in the angular-spectrum representation becomes

$$\mathcal{A}_{\ell p}^C(\mathbf{q}_{\parallel}) \approx \tilde{F}_{\ell p}(q_{\parallel}) e^{i \ell \phi_{q_{\parallel}}} \hat{\varepsilon} - \frac{\sin \theta_q}{\sqrt{2}} \tilde{F}_{\ell p}(q_{\parallel}) e^{i(\sigma + \ell) \phi_{q_{\parallel}}} \hat{\mathbf{z}}. \quad (2.29)$$

This form is consistent with that obtained by Oscar Javier Gomez Sanchez *et al.* in Ref. [30]. Notably, the SOI remains encoded in the phase of the longitudinal component through the  $\sigma + \ell$  dependence, corresponding to the total angular momentum (TAM), as illustrated in Fig. 2.4. However, this phase contribution is generally difficult to detect experimentally, as it tends to average out in intensity-based measurements. To access this TAM contribution, one can employ the interference of two LG beams, forming so-called vector vortex beams (VVBs), which will be discussed in the following section.

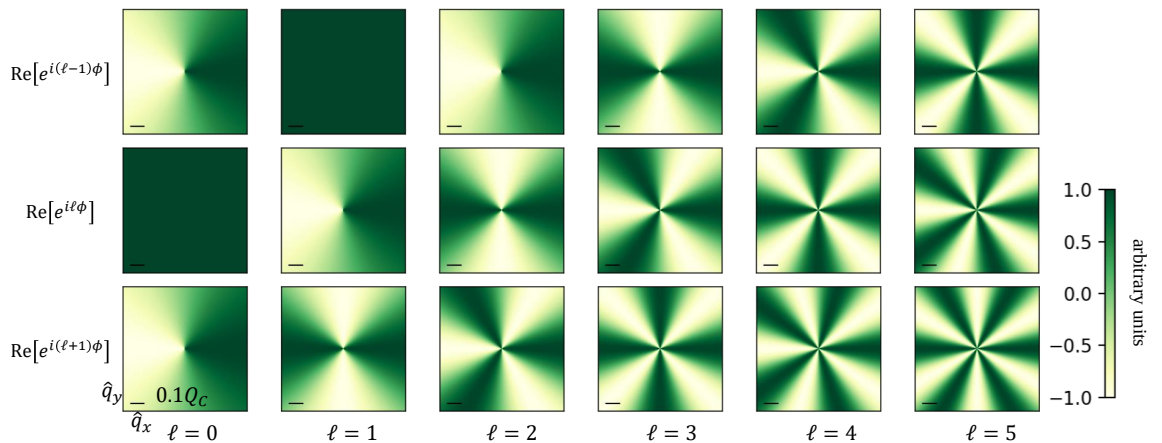


Figure 2.4 : Phase distribution of different components of  $\mathcal{A}_{\ell p}^C(\mathbf{q}_{\parallel})$  for different helicities.

### 2.2.3 Total Angular Momentum in Interference of Twisted Light Beams

In Ref. [30], Oscar Javier Gomez Sanchez *et al.* pointed out that the information of total angular momentum (TAM) encoded in the phase can be lost when calculating exciton transition rates. To retain and control this information, they introduced vector vortex beams (VVBs), which are constructed by superimposing two twisted light beams. The superposition is expressed as

$$\mathcal{A}^{\sigma,\ell,\sigma',\ell'}(\mathbf{q}_{\parallel}; \alpha, \beta) = \cos\left(\frac{\beta}{2}\right) \mathcal{A}^{\sigma,\ell}(\mathbf{q}_{\parallel}) + e^{i\alpha} \sin\left(\frac{\beta}{2}\right) \mathcal{A}^{\sigma',\ell'}(\mathbf{q}_{\parallel}). \quad (2.30)$$

Here, the vector potential  $\mathcal{A}^{\sigma,\ell}(\mathbf{q}_{\parallel}) = \mathcal{A}_{\ell p}^C(\mathbf{q}_{\parallel})$  represents a twisted light mode characterized by SAM ( $\sigma$ ) and OAM ( $\ell$ ). The radial mode index is typically set to  $p = 0$ , since we focus on the interplay between SAM and OAM. The second term corresponds to another twisted light beam carrying a different combination of angular momentum ( $\sigma', \ell'$ ).

The parameter  $\alpha$  controls the relative phase between the two beams, while  $\beta$  determines their relative weight in the superposition. These two parameters can be interpreted as the azimuthal and polar angles on the Poincaré sphere (see ...), providing a convenient geometric representation of the VVB state, which can be written as

$$|\sigma, \ell, \sigma', \ell'; \alpha, \beta\rangle = \cos\left(\frac{\beta}{2}\right) |\sigma, \ell\rangle + e^{i\alpha} \sin\left(\frac{\beta}{2}\right) |\sigma', \ell'\rangle. \quad (2.31)$$

[Add Poincaré sphere figure here]

A VVB contains both transverse and longitudinal components, which can be written as

$$\mathcal{A}^{\sigma,\ell,\sigma',\ell'}(\mathbf{q}_{\parallel}; \alpha, \beta) = \mathcal{A}_{\parallel}^{\sigma,\ell,\sigma',\ell'}(\mathbf{q}_{\parallel}; \alpha, \beta) \hat{\mathbf{e}} + \mathcal{A}_z^{\sigma,\ell,\sigma',\ell'}(\mathbf{q}_{\parallel}; \alpha, \beta) \hat{\mathbf{z}}. \quad (2.32)$$

The squared magnitude of the transverse component is obtained as

$$\begin{aligned} \left| \mathcal{A}_{\parallel}^{\sigma,\ell,\sigma',\ell'}(\mathbf{q}_{\parallel}; \alpha, \beta) \right|^2 &= \cos^2\left(\frac{\beta}{2}\right) \left| \tilde{F}_{\ell p}(q_{\parallel}) \right|^2 + \sin^2\left(\frac{\beta}{2}\right) \left| \tilde{F}_{\ell' p}(q_{\parallel}) \right|^2 \\ &\quad + \frac{1 + \sigma\sigma'}{2} \sin \beta \tilde{F}_{\ell p}(q_{\parallel}) \tilde{F}_{\ell' p}^*(q_{\parallel}) \cos\left(\alpha + \Delta\ell \phi_{\mathbf{q}_{\parallel}}\right). \end{aligned} \quad (2.33)$$

Here,  $\Delta\ell = \ell' - \ell$  is the OAM difference between the two beams. The last term represents the interference between the two beams and vanishes when the two beams carry opposite SAM ( $\sigma' = -\sigma$ ). When  $\ell' = \ell$ , the azimuthal dependence on  $\phi_{\mathbf{q}_{\parallel}}$  disappears, and the interference term becomes independent of the angular coordinate.

[Add figure here]

The squared magnitude of the longitudinal component is obtained as

$$\begin{aligned} \left| \mathcal{A}_z^{\sigma, \ell, \sigma', \ell'}(\mathbf{q}_{\parallel}; \alpha, \beta) \right|^2 &= \frac{\sin^2 \theta_{\mathbf{q}}}{2} \left[ \cos^2 \left( \frac{\beta}{2} \right) \left| \tilde{F}_{\ell_p}(q_{\parallel}) \right|^2 + \sin^2 \left( \frac{\beta}{2} \right) \left| \tilde{F}_{\ell'_p}(q_{\parallel}) \right|^2 \right. \\ &\quad \left. + \sin \beta \tilde{F}_{\ell_p}(q_{\parallel}) \tilde{F}_{\ell'_p}^*(q_{\parallel}) \cos \left( \alpha + \Delta J \phi_{\mathbf{q}_{\parallel}} \right) \right]. \end{aligned} \quad (2.34)$$

Here,  $\Delta J = (\ell' + \sigma') - (\ell + \sigma)$  is the difference in TAM between the two beams. In contrast to the transverse component, the longitudinal component does contain SAM+OAM factor, reflecting its origin from the longitudinal field induced by the transversality condition. The azimuthal dependence disappears when  $\Delta J = 0$ , i.e., when the two beams carry the same total angular momentum.

[Add figure here]

For simplicity, if we consider a VVB carrying opposite SAM ( $\sigma' = -\sigma$ ) and the same OAM ( $\ell' = \ell$ ), the components reduce to

$$\left| \mathcal{A}_{\parallel}^{\sigma, \ell, \sigma', \ell'}(\mathbf{q}_{\parallel}; \alpha, \beta) \right|^2 = \left| \tilde{F}_{\ell_p}(q_{\parallel}) \right|^2, \quad (2.35)$$

$$\left| \mathcal{A}_z^{\sigma, \ell, \sigma', \ell'}(\mathbf{q}_{\parallel}; \alpha, \beta) \right|^2 = \frac{\sin^2 \theta_{\mathbf{q}}}{2} \left| \tilde{F}_{\ell_p}(q_{\parallel}) \right|^2 \left[ 1 + \sin \beta \cos \left( \alpha + \Delta J \phi_{\mathbf{q}_{\parallel}} \right) \right]. \quad (2.36)$$

Where  $\Delta J$  becomes  $\Delta \ell - 2\sigma$ . In this form, the transverse component exhibits an isotropic squared magnitude over the radial direction in momentum space and does not preserve the information of angular momentum. While the longitudinal component reflect the TAM difference between two twisted light.

# Chapter 3 Light-Matter Interaction

## 3.1 Theory of Light–Matter Interaction

### 3.1.1 Optical Transition

In this section, we examine the interaction between light and matter using **Fermi's golden rule**. The transition rate for an optical excitation from the ground state  $|GS\rangle$ , with fully occupied valence bands, to an exciton state  $|\Psi_{S,Q}^X\rangle$  with band index  $S$  and center-of-mass (COM) momentum  $Q$  is given by

$$\Gamma = \frac{2\pi}{\hbar} \left| \langle \Psi_{S,Q}^X | \hat{H}_I | GS \rangle \right|^2 \delta(E_{S,Q}^X - \hbar\omega). \quad (3.1)$$

The delta function enforces energy conservation between the exciton energy  $E_{S,Q}^X$  and the photon energy  $\hbar\omega$ . Within the pseudospin model [6, 30, 38], the exciton state is written as

$$|\Psi_{S,Q}^X\rangle = \frac{1}{\sqrt{\Omega}} \sum_{v\mathbf{k}} \Lambda_{S,Q}(v\mathbf{k}) \hat{c}_{c,\mathbf{k}+Q}^\dagger \hat{h}_{v,-\mathbf{k}}^\dagger |GS\rangle. \quad (3.2)$$

Here, the particle operators  $\hat{c}_{c,\mathbf{k}}^\dagger$  and  $\hat{h}_{v,-\mathbf{k}}^\dagger$  create an electron in the conduction band with momentum  $\mathbf{k}$  and a hole in the valence band with momentum  $-\mathbf{k}$ , respectively. The coefficient  $\Lambda_{S,Q}(v\mathbf{k})$  represents the amplitude of the electron–hole configuration for exciton band  $S$ , and  $\Omega$  is the area of the transition-metal dichalcogenide (TMD) monolayer. The matrix element in Eq. (3.1) can be expressed, within the **rotating wave approximation** [39], as

$$\langle \Psi_{S,Q}^X | \hat{H}_I | GS \rangle = \frac{1}{\sqrt{\Omega}} \sum_{v\mathbf{k}} \Lambda_{S,Q}^*(v\mathbf{k}) \langle \psi_{c,\mathbf{k}+Q} | \hat{H}_I | \psi_{v,\mathbf{k}} \rangle, \quad (3.3)$$

where, for weak fields, the light–matter interaction Hamiltonian is given by [6, 30]

$$\hat{H}_I = \frac{|e|\hbar}{2m_e} \mathbf{A}(\mathbf{r}, t) \cdot \mathbf{p},$$

with  $\mathbf{A}(\mathbf{r}, t)$  the vector potential in the **Coulomb gauge**,  $\mathbf{p}$  the momentum operator, and  $m_e$  ( $e$ ) the electron mass (charge). If the vector potential is expanded in a **plane-wave basis** as in

Eq. (2.25), the matrix element in Eq. (3.3) becomes

$$\begin{aligned}\langle \psi_{c,k+Q} | \hat{H}_I | \psi_{v,k} \rangle &= \frac{|e|}{2m_e} \sum_{q_{\parallel}} \mathcal{A}(q_{\parallel}) [\hat{\epsilon} \cdot \langle \psi_{c,k+Q} | e^{iq_{\parallel} \cdot \mathbf{r}} \hat{\mathbf{p}} | \psi_{v,k} \rangle] e^{i\omega t} \\ &\approx \frac{|e|}{2m_e} \sum_{q_{\parallel}} \mathcal{A}(q_{\parallel}) [\hat{\epsilon} \cdot \delta_{q_{\parallel},Q} \langle \psi_{c,k} | \hat{\mathbf{p}} | \psi_{v,k} \rangle] e^{i\omega t},\end{aligned}\quad (3.4)$$

where the approximation follows from the **electric dipole approximation**. In this approximation, the Bloch wave functions are written as  $\psi_{n,k}(\mathbf{r}) = u_{n,k}(\mathbf{r})e^{i\mathbf{k} \cdot \mathbf{r}}$ , and the plane-wave factors combine to yield the Kronecker delta  $\delta_{q_{\parallel},Q}$ , enforcing momentum conservation between the in-plane photon wavevector and the exciton COM momentum. Substituting this result back, the matrix element in Eq. (3.1) [6, 30] becomes

$$\langle \Psi_{S,Q}^X | \hat{H}_I | GS \rangle = \frac{E_g}{2i\hbar} \mathcal{A}(Q) (\hat{\epsilon} \cdot \mathbf{D}_{S,Q}^{X*}) e^{i\omega t}, \quad (3.5)$$

where  $\mathbf{D}_{S,Q}^X$  is the exciton transition dipole moment and  $E_g$  is the band gap at the  $K/K'$  valleys. This expression shows that optical excitation is governed by the alignment between the light polarization and the transition dipole moment. The transition rate is therefore proportional to both the magnitude of the transition dipole moment and the field amplitude. In terms of the defined optical matrix element,

$$\begin{aligned}\tilde{M}_{S,Q}^{\sigma,\ell,\sigma',\ell'} &\equiv \frac{1}{\sqrt{\Omega}} \sum_{vck} \Lambda_{S,Q}^*(vck) \left\langle \psi_{c,k+Q} \left| \frac{|e|}{2m_e} \mathcal{A}_{\parallel}^{\sigma,\ell,\sigma',\ell'}(Q; \alpha, \beta) \cdot \mathbf{p} \right| \psi_{v,k} \right\rangle \\ &\approx \frac{E_g}{2i\hbar} \mathcal{A}_{\parallel}^{\sigma,\ell,\sigma',\ell'}(Q; \alpha, \beta) (\hat{\epsilon} \cdot \mathbf{D}_{S,Q}^{X*}).\end{aligned}\quad (3.6)$$

Fermi's golden rule can then be written as [30]

$$\Gamma_{S,Q}^{\sigma,\ell,\sigma',\ell'} = \frac{2\pi}{\hbar} \left| \tilde{M}_{S,Q}^{\sigma,\ell,\sigma',\ell'} \right|^2 \rho(\hbar\omega = E_{S,Q}^X) \propto \left| \mathcal{A}_{\parallel}^{\sigma,\ell,\sigma',\ell'}(Q; \alpha, \beta) (\hat{\epsilon} \cdot \mathbf{D}_{S,Q}^X) \right|^2. \quad (3.7)$$

The transition dipole moment, obtained from density functional theory (DFT) combined with the Bethe–Salpeter equation (BSE), is given in the small- $Q$  limit by [6, 30]:

#### Valley-mixed excitons

$$\begin{aligned}\text{Valley excitons} \quad \mathbf{D}_{K,Q}^X &= \frac{D_0^X}{\sqrt{2}} (\hat{x} + i\hat{y}), \\ \mathbf{D}_{K',Q}^X &= \frac{D_0^X}{\sqrt{2}} (\hat{x} - i\hat{y})\end{aligned}\quad (3.8)$$

$$\begin{aligned}\mathbf{D}_{+,Q}^X &= \frac{D_0^X}{\sqrt{2}} (\cos \phi_Q \hat{x} + \sin \phi_Q \hat{y}), \\ \mathbf{D}_{-,Q}^X &= \frac{iD_0^X}{\sqrt{2}} (-\sin \phi_Q \hat{x} + \cos \phi_Q \hat{y})\end{aligned}\quad (3.9)$$

#### Gray excitons

$$\mathbf{D}_{G,Q}^X = D_{G,0}^X \hat{z}. \quad (3.10)$$

Here, the symbols  $K$ ,  $K'$ ,  $+$ ,  $-$ , and  $G$  denote the  $K$  valley,  $K'$  valley, upper band, lower band, and gray exciton, respectively. The magnitude of the dipole moment of BXs (GX) is  $D_0^X$

( $D_{G,0}^X$ ) From Eq. (3.7), it becomes clear why a single twisted-light beam, whose TAM is encoded purely in its phase, does not produce a transition rate that reflects SOI or TAM. This is because the transition rate depends on the squared magnitude of the vector potential, which removes all phase information.

Based on the above formulation, the total transition rate is obtained by summing over all contributions in momentum space. This allows for a systematic evaluation of SOI effects for different combinations of spin angular momentum (SAM) and orbital angular momentum (OAM) in VVBs:

$$\Gamma_S^{\sigma,\ell,\sigma',\ell'} = \sum_Q \Gamma_{S,Q}^{\sigma,\ell,\sigma',\ell'}. \quad (3.11)$$

## 3.2 Optical Transition with Vector Vortex Beam in TMD

### 3.2.1 SOI of Transition Rate in Momentum Space

In comparison between valley-mixed excitons and gray excitons, the  $\phi_Q$  dependence of the valley-mixed exciton transition dipoles in Eq. (3.9) provides an alternative pathway for SOI. This angular dependence allows the spin index  $\sigma$  to enter the phase of the optical matrix element, thereby enabling valley-mixed excitons to exhibit SOI features analogous to those of gray excitons, both under single twisted-light excitation and in VVBs.

In contrast, for gray excitons, the SOI manifests directly through the interference term in Eq. (2.34). This is because the gray-exciton transition dipole is oriented along the out-of-plane ( $z$ ) direction and is isotropic in the in-plane coordinates, making the interference-induced contribution the dominant mechanism for revealing SOI.

For comparison, we now focus on how the transverse component of VVBs excites valley-mixed excitons. Substituting Eq. (2.33) into

### 3.2.2 Total Transition Rate with Total Angular Momentum

# Chapter 4    Exciton Wave Packet States

## 4.1    Exciton Wave Packet States



## Chapter 5 Summary and outlook



# References

- [1] Nicolas Roisin et al. “Phonon-limited mobility for electrons and holes in highly-strained silicon”. In: *npj Computational Materials* 10.1 (2024), p. 242.
- [2] Aaron Bostwick et al. “Quasiparticle dynamics in graphene”. In: *Nature Physics* 3.1 (2007), pp. 36–40.
- [3] Andrea Splendiani et al. “Emerging photoluminescence in monolayer MoS<sub>2</sub>”. In: *Nano Letters* 10.4 (2010), pp. 1271–1275.
- [4] Alexander Krasnok and Andrea Alù. “Valley-selective response of nanostructures coupled to 2D transition-metal dichalcogenides”. In: *Applied Sciences* 8.7 (2018), p. 1157. DOI: 10.3390/app8071157.
- [5] Guan-Hao Peng et al. “Distinctive Signatures of the Spin- and Momentum-Forbidden Dark Exciton States in the Photoluminescence of Strained WSe<sub>2</sub> Monolayers under Thermalization”. In: *Nano Letters* 19.4 (2019), pp. 2299–2307. DOI: 10.1021/acs.nanolett.8b04713.
- [6] Guan-Hao Peng et al. “Tailoring the superposition of finite-momentum valley exciton states in transition-metal dichalcogenide monolayers by using polarized twisted light”. In: *Physical Review B* 105.11 (2022), p. 115421. DOI: 10.1103/PhysRevB.105.115421.
- [7] Lorenz Maximilian Schneider et al. “Direct measurement of the radiative pattern of bright and dark excitons and exciton complexes in encapsulated tungsten diselenide”. In: *Scientific Reports* 10.1 (2020), p. 8091. DOI: 10.1103/s41598-020-65091-x.
- [8] J. H. Poynting. “The Wave Motion of a Revolving Shaft, and a Suggestion as to the Angular Momentum in a Beam of Circularly Polarised Light”. In: *Proceedings of the Royal Society of London A: Mathematical, Physical and Engineering Sciences* 82 (1909), pp. 560–567.
- [9] R. A. Beth. “Mechanical Detection and Measurement of the Angular Momentum of Light”. In: *Physical Review* 50 (1936), pp. 115–125.
- [10] L. Allen et al. “Orbital angular momentum of light and the transformation of Laguerre-Gaussian laser modes”. In: *Physical Review A* 45 (1992), pp. 8185–8189.
- [11] Josef-Maria Jauch and Fritz Rohrlich. *The Theory of Photons and Electrons: The Relativistic Quantum Field Theory of Charged Particles with Spin One-Half*. 2nd ed. Berlin: Springer-Verlag, 1976, p. 34. DOI: 10.1007/978-3-642-80951-4.
- [12] Andrew Forbes, Light Mkhumbuza, and Liang Feng. “Orbital angular momentum lasers”. In: *Nature Reviews Physics* (May 2024). DOI: 10.1038/s42254-024-00715-2. URL: <https://doi.org/10.1038/s42254-024-00715-2>.

- [13] Manuel Erhard et al. “Twisted photons: new quantum perspectives in high dimensions”. In: *Light: Science & Applications* 7.3 (2018), pp. 17146–17146. DOI: 10.1038/lsa.2017.146.
- [14] H He et al. “Direct observation of transfer of angular momentum to absorptive particles from a laser beam with a phase singularity”. In: *Physical Review Letters* 75.5 (1995), pp. 826–829.
- [15] L Paterson et al. “Controlled rotation of optically trapped microscopic particles”. In: *Science* 292.5518 (2001), pp. 912–914. DOI: 10.1126/science.1058591.
- [16] Alexander B Stilgoe, Timo A Nieminen, and Halina Rubinsztein-Dunlop. “Controlled transfer of transverse orbital angular momentum to optically trapped birefringent microparticles”. In: *Nature Photonics* 16.5 (2022), pp. 346–351. DOI: 10.1038/s41566-022-00973-x.
- [17] Graham Gibson et al. “Free-space information transfer using light beams carrying orbital angular momentum”. In: *Optics Express* 12.22 (2004), pp. 5448–5456.
- [18] Jian Wang et al. “Terabit free-space data transmission employing orbital angular momentum multiplexing”. In: *Nature Photonics* 6.7 (2012), pp. 488–496.
- [19] Nenad Bozinovic et al. “Terabit-scale orbital angular momentum mode division multiplexing in fibers”. In: *Science* 340.6140 (2013), pp. 1545–1548.
- [20] Giuseppe Vallone et al. “Free-Space Quantum Key Distribution by Rotation-Invariant Twisted Photons”. In: *Physical Review Letters* 113.6 (2014), p. 060503. DOI: 10.1103/PhysRevLett.113.060503.
- [21] Alicia Sit et al. “High-dimensional intracity quantum cryptography with structured photons”. In: *Optica* 4.9 (2017), pp. 1006–1010. DOI: 10.1364/OPTICA.4.01006.
- [22] Fabrizio Tamburini et al. “Twisting of light around rotating black holes”. In: *Nature Physics* 7.3 (2011), pp. 195–197.
- [23] Dong-Sheng Ding et al. “Single-photon-level quantum image memory based on cold atomic ensembles”. In: *Nature Communications* 4 (2013), p. 2527.
- [24] Alice Nicolas et al. “A quantum memory for orbital angular momentum photonic qubits”. In: *Nature Photonics* 8 (2014), pp. 234–238.
- [25] Lluís Torner, Juan P Torres, and Silvia Carrasco. “Digital spiral imaging”. In: *Optics Express* 13.3 (2005), pp. 873–881.
- [26] Nestor Uribe-Patarroyo et al. “Object identification using correlated orbital angular momentum states”. In: *Physical Review Letters* 110.4 (2013), p. 043601.
- [27] Miles Padgett and Richard Bowman. “Tweezers with a twist”. In: *Nature Photonics* 5.6 (2011), pp. 343–348.
- [28] Konstantin Y Bliokh et al. “Spin–orbit interactions of light”. In: *Nature Photonics* 9.12 (2015), pp. 796–808. DOI: 10.1038/nphoton.2015.201.
- [29] Timo A Nieminen et al. “Angular momentum of a strongly focused Gaussian beam”. In: *Journal of Optics A: Pure and Applied Optics* 10.11 (2008), p. 115005. DOI: 10.1088/1464-4258/10/11/115005.

- [30] Oscar Javier Gomez Sanchez et al. “Enhanced Photo-excitation and Angular-Momentum Imprint of Gray Excitons in WSe<sub>2</sub> Monolayers by Spin – Orbit-Coupled Vector Vortex Beams”. In: *ACS Nano* 18.4 (2024), pp. 3214–3224. DOI: 10.1021/acsnano.3c10336.
- [31] Konstantin Y Bliokh and Franco Nori. “Transverse and longitudinal angular momenta of light”. In: *Physics Reports* 592 (2015), pp. 1–38. DOI: 10.1016/j.physrep.2015.06.003.
- [32] Stephen M Barnett and Les Allen. “Orbital angular momentum and nonparaxial light beams”. In: *Optics Communications* 110.5-6 (1994), pp. 670–678. DOI: 10.1016/0030-4018(94)90269-0.
- [33] David L Andrews and Mohamed Babiker, eds. *The Angular Momentum of Light*. Cambridge: Cambridge University Press, 2012, p. 12. DOI: 10.1017/CBO9780511795213.
- [34] LC Dávila Romero, DL Andrews, and M Babiker. “A quantum electrodynamics framework for the nonlinear optics of twisted beams”. In: *Journal of Optics B: Quantum and Semiclassical Optics* 4.2 (2002), S66–S72. DOI: 10.1088/1464-4266/4/2/361.
- [35] Kayn A Forbes, Dale Green, and Garth A Jones. “Relevance of longitudinal fields of paraxial optical vortices”. In: *Journal of Optics* 23.7 (2021), p. 075401. DOI: 10.1088/2040-8986/ac0411.
- [36] VV Klimov et al. “Mapping of focused Laguerre-Gauss beams: The interplay between spin and orbital angular momentum and its dependence on detector characteristics”. In: *Physical Review A* 85.5 (2012), p. 053834. DOI: 10.1103/PhysRevA.85.053834.
- [37] Shojiro Nemoto. “Nonparaxial Gaussian beams”. In: *Applied Optics* 29.13 (1990), pp. 1940–1946. DOI: 10.1364/AO.29.001940.
- [38] Kristan Bryan Simbulan et al. “Selective Photoexcitation of Finite-Momentum Excitons in Monolayer MoS<sub>2</sub> by Twisted Light”. In: *ACS Nano* 15.2 (2021), pp. 3117–3126. DOI: 10.1021/acsnano.0c09772.
- [39] David J Griffiths and Darrell F Schroeter. *Introduction to Quantum Mechanics*. 3rd ed. Cambridge: Cambridge University Press, 2018, p. 410.

# Appendix

## Appendix.1 Wave equations from Maxwell's equations in free space

In the electrodynamics framework, Maxwell's equations can be written as

$$\nabla \cdot \mathbf{E}(\mathbf{r}, t) = \frac{\rho}{\epsilon_0} = 0, \quad (\text{A.1.1})$$

$$\nabla \cdot \mathbf{B}(\mathbf{r}, t) = 0, \quad (\text{A.1.2})$$

$$\nabla \times \mathbf{E}(\mathbf{r}, t) = -\frac{\partial}{\partial t} \mathbf{B}(\mathbf{r}, t), \quad (\text{A.1.3})$$

$$\nabla \times \mathbf{B}(\mathbf{r}, t) = \mu_0 \mathbf{J} + \frac{1}{c^2} \frac{\partial}{\partial t} \mathbf{E}(\mathbf{r}, t) = \frac{1}{c^2} \frac{\partial}{\partial t} \mathbf{E}(\mathbf{r}, t). \quad (\text{A.1.4})$$

Here  $\nabla \equiv \frac{\partial}{\partial x} \hat{\mathbf{x}} + \frac{\partial}{\partial y} \hat{\mathbf{y}} + \frac{\partial}{\partial z} \hat{\mathbf{z}}$  represents the gradient operator along the three spatial coordinates. These equations can also be expressed in terms of electromagnetic potentials, which will be useful later when discussing light-matter interaction. First, the divergence-free condition of Eq. (A.1.2) allows the magnetic field to be written in terms of a **vector potential**  $\mathbf{A}(\mathbf{r}, t)$  as

$$\mathbf{B}(\mathbf{r}, t) = \nabla \times \mathbf{A}(\mathbf{r}, t). \quad (\text{A.1.5})$$

Substituting this expression into Faraday's law (Eq. (A.1.3)) yields a quantity whose curl vanishes. This allows the introduction of a **scalar potential**  $\phi(\mathbf{r}, t)$  such that

$$\nabla \phi(\mathbf{r}, t) = \left( \mathbf{E}(\mathbf{r}, t) + \frac{\partial}{\partial t} \mathbf{A}(\mathbf{r}, t) \right). \quad (\text{A.1.6})$$

Therefore, the electric field can be expressed in terms of the scalar and vector potentials as

$$\mathbf{E}(\mathbf{r}, t) = -\nabla \phi(\mathbf{r}, t) - \frac{\partial}{\partial t} \mathbf{A}(\mathbf{r}, t). \quad (\text{A.1.7})$$

This expression automatically satisfies both Eq. (A.1.2) and Faraday's law (Eq. (A.1.3)). Finally, Gauss's law (Eq. (A.1.1)) and Ampère's law with Maxwell's correction (Eq. (A.1.4)) can be rewritten as two coupled wave equations for the electromagnetic potentials:

$$\nabla^2 \phi(\mathbf{r}, t) + \frac{\partial}{\partial t} (\nabla \cdot \mathbf{A}(\mathbf{r}, t)) = 0 \quad (\text{A.1.8})$$

$$\nabla^2 \mathbf{A}(\mathbf{r}, t) - \frac{1}{c^2} \frac{\partial^2}{\partial t^2} \mathbf{A}(\mathbf{r}, t) - \nabla \left( \nabla \cdot \mathbf{A}(\mathbf{r}, t) + \frac{1}{c^2} \frac{\partial}{\partial t} \phi(\mathbf{r}, t) \right) = 0. \quad (\text{A.1.9})$$

## Appendix.2 scalar Helmholtz equation

We begin with the Lorenz gauge. First, we assume a **monochromatic wave** ansatz,  $\mathbf{A}^L(\mathbf{r}, t) = \text{Re} [\mathbf{A}^L(\mathbf{r})e^{-i\omega t}]$ , which transforms Eq. (2.8) into the vector Helmholtz equation

$$\nabla^2 \mathbf{A}^L(\mathbf{r}) + q_0^2 \mathbf{A}^L(\mathbf{r}) = 0. \quad (\text{A.2.1})$$

Next, we introduce the **transverse-wave** ansatz,  $\mathbf{A}^L(\mathbf{r}) = \hat{\varepsilon} A^L(\mathbf{r})$ , where  $\hat{\varepsilon}$  is a polarization unit vector. Substituting this form into Eq. (A.2.1) reduces the equation to the scalar Helmholtz equation

$$\nabla^2 A^L(\mathbf{r}) + q_0^2 A^L(\mathbf{r}) = 0. \quad (\text{A.2.2})$$

